Methodological Considerations in Clinical Exome Sequencing for Recessive Disorders

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1. Segregation analysis — section 1

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 8050610, 68103859, 50015914.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 34210829, 12960758, 8050610.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 12513232, 68103859, 47459230.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 55137901, 89300592, 6905075.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 68103859, 20767632, 63655258.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 8050610, 96779226, 56765723.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 41936281, 12244083, 89927758.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 50237932, 50015914, 47459230.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Founder alleles complicate carrier-frequency interpretation in population isolates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 99159620, 11457510, 74674916.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 56765723, 12960758, 11457510.

2. Population-frequency databases — section 2

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 47459230, 50237932, 56765723.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 68103859, 47459230, 89300592.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 57965585, 88145754, 6905075.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 56765723, 28553419, 47459230.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 50015914, 44244434, 57965585.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 89927758, 89927758, 12513232.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 12960758, 20767632, 12513232.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 68103859, 88145754, 75387967.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 96779226, 75374952, 8050610.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 75374952, 6905075, 96779226.

3. DNA isolation and quality control — section 3

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 88145754, 20767632, 41936281.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 68103859, 99937139, 75387967.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 96779226, 9010580, 44244434.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 96779226, 96779226, 47197252.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 55137901, 89300592, 44244434.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 9010580, 77567010, 55137901.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 28553419, 12513232, 89927758.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 77567010, 55137901, 44244434.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 34210829, 16405705, 75374952.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Founder alleles complicate carrier-frequency interpretation in population isolates. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 50015914, 11457510, 12513232.

Table D3. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	10456922	4972567	0.8437
2	13674883	8527619	0.7703
3	99239119	8843734	0.0387
4	45299748	2208638	0.5971
5	20803455	2732056	0.1815
6	93092167	2787286	0.7207
7	5964298	6903063	0.3646
8	96681154	3995095	0.9686

4. Ethnicity matching — section 4

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 38605065, 99937139, 16405705.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 41936281, 47197252, 89300592.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Founder alleles complicate carrier-frequency interpretation in population isolates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 38605065, 16405705, 77567010.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 57965585, 8050610, 96779226.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 96779226, 89927758, 34210829.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 17463900, 55137901, 63655258.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 44244434, 12513232, 11457510.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 96779226, 88145754, 89269366.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 77567010, 20767632, 47197252.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 12244083, 20486949, 17463900.

5. Population-frequency databases — section 5

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 20767632, 13905132, 34210829.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 56765723, 89269366, 88145754.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 47197252, 16405705, 92183658.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 55137901, 38605065, 50237932.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 92183658, 89927758, 17463900.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 89927758, 13905132, 44244434.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 92183658, 6905075, 50237932.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Founder alleles complicate carrier-frequency interpretation in population isolates. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 75387967, 75374952, 41936281.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 99937139, 78714110, 89927758.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 56765723, 47459230, 96779226.

6. Population-frequency databases — section 6

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 47197252, 8050610, 63655258.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 75387967, 9010580, 63655258.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 57965585, 96779226, 89927758.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 20486949, 92183658, 47459230.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 20767632, 96779226, 50237932.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 55137901, 68103859, 50237932.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 96779226, 92183658, 68103859.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 8050610, 17463900, 12244083.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 77567010, 12244083, 74674916.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Founder alleles complicate carrier-frequency interpretation in population isolates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 99937139, 12244083, 11457510.

Table D6. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	10039096	7594017	0.1733
2	95846886	7477726	0.9963
3	91626995	5371793	0.6675
4	95332912	1099496	0.5499
5	39071765	5042437	0.8507
6	95637400	2930762	0.7931
7	68384040	3772150	0.1221
8	27075128	2339967	0.2373

7. Segregation analysis — section 7

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 77567010, 11457510, 88145754.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 47197252, 89300592, 89300592.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 96779226, 6905075, 47197252.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 56765723, 41936281, 20486949.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 57965585, 75387967, 50015914.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 99937139, 77567010, 8050610.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 28553419, 28553419, 12513232.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 50237932, 68103859, 74674916.

Founder alleles complicate carrier-frequency interpretation in population isolates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 13905132, 78714110, 75374952.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 92183658, 20486949, 57965585.

8. Segregation analysis — section 8

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Founder alleles complicate carrier-frequency interpretation in population isolates. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 20767632, 12513232, 75387967.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 41936281, 77567010, 9010580.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 96779226, 17463900, 12513232.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 17463900, 57965585, 75387967.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 41936281, 6905075, 80585778.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 41936281, 6905075, 17463900.

Founder alleles complicate carrier-frequency interpretation in population isolates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 78714110, 80585778, 20767632.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 56765723, 11457510, 74674916.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 20486949, 68103859, 77567010.

Founder alleles complicate carrier-frequency interpretation in population isolates. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 55137901, 41936281, 96779226.

9. Counseling implications — section 9

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 47459230, 16405705, 44244434.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Founder alleles complicate carrier-frequency interpretation in population isolates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 41936281, 50237932, 50237932.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 12513232, 20767632, 41936281.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 88145754, 75374952, 28553419.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 38605065, 34210829, 16405705.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 20767632, 38605065, 74674916.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 57965585, 47197252, 88145754.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 99937139, 34210829, 96779226.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 89300592, 41936281, 88145754.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 77567010, 75387967, 38605065.

Table D9. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	70628349	6332070	0.3141
2	23344151	7721874	0.9113
3	46046354	9174943	0.3546
4	96860856	4445261	0.6099
5	25905259	4145904	0.2794
6	40163879	3782642	0.9406
7	40045150	4858684	0.7045
8	92742432	8214110	0.3174

10. Annotation pipelines — section 10

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 47459230, 89927758, 12513232.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 96779226, 74674916, 38605065.

Founder alleles complicate carrier-frequency interpretation in population isolates. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 11457510, 9010580, 88145754.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 50015914, 20486949, 9010580.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 89269366, 77567010, 57965585.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Founder alleles complicate carrier-frequency interpretation in population isolates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 75387967, 78714110, 20486949.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 89300592, 12960758, 50237932.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 89269366, 75374952, 68103859.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 13905132, 47197252, 80585778.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 12244083, 44244434, 57965585.

11. Reporting standards (ACMG) — section 11

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 78714110, 68103859, 99937139.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 63655258, 47459230, 9010580.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 89269366, 74674916, 55137901.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 28553419, 68103859, 89269366.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 6905075, 75374952, 57965585.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 16405705, 88145754, 50015914.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 20767632, 16405705, 89300592.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Founder alleles complicate carrier-frequency interpretation in population isolates. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 20767632, 63655258, 47197252.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 20486949, 77567010, 41936281.

Founder alleles complicate carrier-frequency interpretation in population isolates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 75387967, 47197252, 96779226.

12. Library preparation — section 12

Founder alleles complicate carrier-frequency interpretation in population isolates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 20486949, 6905075, 11457510.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 75374952, 34210829, 47459230.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Founder alleles complicate carrier-frequency interpretation in population isolates. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 56765723, 89927758, 92183658.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 78714110, 89300592, 12960758.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 55137901, 55137901, 63655258.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 13905132, 47459230, 47197252.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 6905075, 13905132, 56765723.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 89927758, 12513232, 8050610.

Founder alleles complicate carrier-frequency interpretation in population isolates. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 17463900, 99159620, 92183658.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 50237932, 47459230, 96779226.

Table D12. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	31552485	9065666	0.2178
2	7675790	2361084	0.5039
3	38924283	3938578	0.8152
4	98282614	9675527	0.3193

Index	Metric A	Metric B	Metric C
5	80351151	5389867	0.2374
6	19328089	8755141	0.2216
7	40432555	4626316	0.0618
8	79512216	2948247	0.6257

13. Population-frequency databases — section 13

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 47197252, 75387967, 89300592.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 13905132, 38605065, 17463900.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 77567010, 89300592, 89300592.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 74674916, 99159620, 44244434.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 20486949, 8050610, 55137901.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Founder alleles complicate carrier-frequency interpretation in population isolates. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 38605065, 99159620, 44244434.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Founder alleles complicate carrier-frequency interpretation in population isolates. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 38605065, 17463900, 20767632.

Founder alleles complicate carrier-frequency interpretation in population isolates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 55137901, 6905075, 13905132.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 6905075, 9010580, 38605065.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 50237932, 68103859, 50015914.

14. Reporting standards (ACMG) — section 14

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 12513232, 99937139, 44244434.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 74674916, 80585778, 80585778.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 13905132, 80585778, 12960758.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 20486949, 89269366, 89300592.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 99159620, 41936281, 6905075.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 89269366, 13905132, 77567010.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 13905132, 78714110, 34210829.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 47197252, 12513232, 17463900.

Founder alleles complicate carrier-frequency interpretation in population isolates. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 47459230, 63655258, 16405705.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 63655258, 50237932, 63655258.

End of distractor document. No content above answers any specific pedigree-level question; this document is included to test whether downstream agents correctly select the relevant source paper.